108 Language-Centered World Simulator for Testing Various Approaches to Machine Consciousness

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Most of current theories of consciousness underline the importance of embodiment as the base of any kind of first-person experience. However, we assume that a hypothetical person with all senses disabled who communicates with the outside world through a neurological input/output natural language device would be able to reason about the outside world and to keep her/his free will. It would be easy to manipulate such person if all the communication is through third persons but with more sophisticated I/O devices she/he could also receive visual, haptic or audio inputs and be able to confirm if e.g. requests were fulfilled or not. We assume that majority of population would consider such person as a conscious being, especially in the "cyborg" phase, when the dysfunctional biological sensors are replaced with artificial ones. With this assumption (cyborgs are not zombies) in mind, we propose a real world simulator that is built (by now) only with natural language, a proving ground that could be used as the first phase for a Turing test for consciousness. Reasoning behind the idea: 1) because technology is not ready for the human-level cognition, engineers struggle to test or compare their algorithms when it comes to natural, responsible or moral decision making; 2) to make their reasoning understandable for humans, machines need to explain their motivations and decisions in natural language - the textual simulator requires language input and output which is easy to process and simplifies comparison between participating systems; 3) built automatically from the Internet resources, the simulated worlds can differ (e.g. depending on language) or be randomly chosen without loosing statistically general character. The proposed World Simulator, currently used for learning different experiences by artificial agents to provide variety of artificial personalities, consists of following components: a) environment builder which uses web-mining to gather data about places and to place objects and persons inside; b) five senses classifier represented by adjective groups related to a given sense (e.g. hard, soft, wet, dry, etc. for "touch"), which allows an agent to perceive objects "as already grounded" in natural language instead of raw signals from real sensors; c) physical situation builder which calculates possible changes of objects' positions or characteristics; d) social situation builder which retrieves possible scenarios of human / animal behavior and their consequences. Builders utilize common sense processing and are prone to errors, but a tested agent should be able to recognize not only abnormalities inside itself but also cognition errors and possibilities of positive or negative influences / consequences outside. Decisions made inside such simple virtual world are safer than ones in the real world, and easier to judge by people because the situations are taken from the real world descriptions, not simple game scenarios usually used for testing cognitive architectures. Although the simulator unifies ununifiable qualia (redness is one adjective "red"), language-centered approach should simplify comparing human and non-human conscious behavior with what Steel calls "inner voice" on both human's and machine's sides.

109 Subjectivity, Objectivity, Feelings, and Vision.

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If Physicalism is a viable approach to consciousness, we must address two basic problems: 1) the relationship between first-person subjective experience and third-person objective description; and 2) the source of subjective 'feelings' such as pain, touch, etc. and even visual experience. In particular, we argue that: 1) Reported first-person and third-person viewpoints of an object can be analytically described in terms of one another in the limit that the third-person viewpoint is constructed of a sufficient number of first-person reported viewpoints; and 2) Subjective 'feelings' can be analogously described in terms of off-sets between current sensory data and built-in set-points that are constitutive of a Very Clever Robot (VCR). For conscious beings, we suggest that 'feelings' are 'off-sets' that have overflowed into access conscious states (Block (2011)) via a Reflection Principle (described below) operating on input sensory data. Thus they can be reported and acting on. Finally, as an example, we apply our conclusions to subjective visual experience for which 'feeling' is of different character than the feeling of pain, touch. The Reflection Principle: a) Inputs to the senses from the external/internal world (eye, tactile, etc.) are encoded into neural spikes that feed ingoing into the brain through many serial and parallel modules. These modules process the neural signals and build various neural representations of the world. For each sense, there might be many representations, each representing some aspect of the input sensory data - what, where, color, location, etc.; b) Neural projection